

MID SEM. EXAMINATION, Sep.-2025

Course Code: ECECC304/EIECC304

Course Title: Microelectronics Circuits and Applications

Time: 01:30 Hrs.

Max. Marks: 20

Note: Attempt all questions. All questions carry equal marks. Missing data/information if any must be suitably assumed and mentioned in the answer.

Q. No.	Question	Marks	CO
1	(a) Define Common-Mode Rejection Ratio (CMRR) and explain how it can be improved. (b) The differential amplifier shown in Fig. 1 uses transistors having $\beta = 200$ and Early voltage $V_A = \infty$, biased at $I_{EE} = 100 \mu A$. Determine R_C and R_E such that $A_{vM} = 400$ and $CMRR = 90$ dB.	1+3	CO1
2	(a) Draw the complete circuit of a MOSFET-based cascode amplifier and draw its small-signal equivalent circuit. (b) Find the output resistance and voltage gain of MOSFET-based Cascode amplifier drawn in (a) part.	1+3	CO1
3	(a) Draw the small equivalent circuit of the multistage amplifier given in Fig. 2.	1+3	CO1

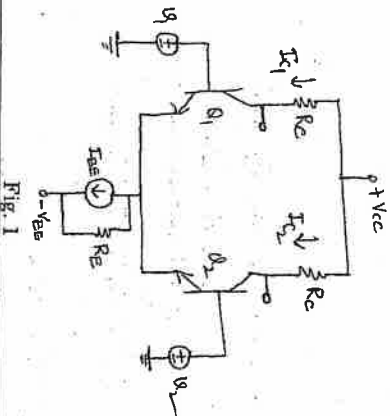


Fig. 1

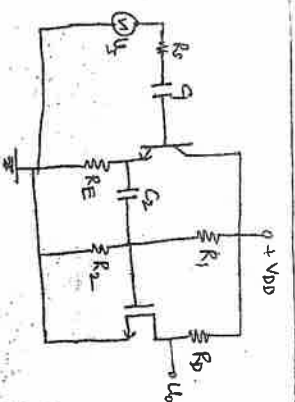


Fig. 2

(b) Find input resistance and overall voltage gain of the multistage amplifier shown in Fig. 2.

(a) State and prove Miller's theorem.
(b) For the circuit shown in Fig. 3, find voltage gain with given $R_D = 10 \text{ k}\Omega$, $R_S = 1 \text{ k}\Omega$, $R_F = 50 \text{ k}\Omega$, $r_o = 50 \text{ k}\Omega$, and $g_m = 1 \text{ mS}$.

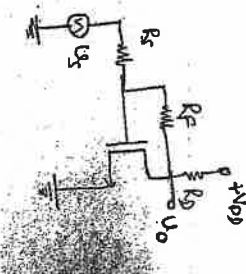


Fig. 3

(a) Draw the high-frequency small signal equivalent circuit model of a 4-terminal MOSFET.

(b) Find the high frequency expression for the dominant pole using the time constant method for common emitter circuit given in Fig. 4.

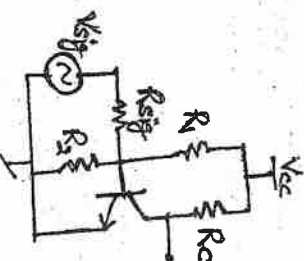


Fig. 4

4	(a) State and prove Miller's theorem. (b) For the circuit shown in Fig. 3, find voltage gain with given $R_D = 10 \text{ k}\Omega$, $R_S = 1 \text{ k}\Omega$, $R_F = 50 \text{ k}\Omega$, $r_o = 50 \text{ k}\Omega$, and $g_m = 1 \text{ mS}$.	1+3	CO1
5	(a) Draw the high-frequency small signal equivalent circuit model of a 4-terminal MOSFET. (b) Find the high frequency expression for the dominant pole using the time constant method for common emitter circuit given in Fig. 4.	1+3	CO2